

QuickVoice Master Admin Configuration Guide

Comprehensive Enterprise Handbook for Multi-Provider Telephony (Vobiz, Twilio, Telnyx), Vernacular AI (Sarvam, Deepgram), Cloud LLM Matrix (OpenRouter, DeepSeek, OpenAI, Claude 3.7, Gemini), LiveKit WebRTC, and Admin Role Security

Product Version:	QuickVoice Enterprise v2.5
Access Restriction:	Admin & Owner Accounts Only
Telephony Providers:	Vobiz SIP (India 140/92), Twilio, Telnyx, LiveKit
Speech & Voice Providers:	Sarvam AI (Saaras/Bulbul), Deepgram, ElevenLabs
Cloud LLM Engines:	OpenRouter, DeepSeek R1/V3, Claude 3.7, GPT-4o, Gemini 2.0
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1. Architecture Overview & Master Secrets

1.1 System Components & Environment Topography

The QuickVoice ecosystem operates across three synchronized execution planes:

- 1. **API Server (Node.js / Express / Prisma):** Manages multi-tenant authentication, organization hierarchy, role-based access control, billing wallets, SIP routing metadata, and webhook dispatches. Configured via `apps/server/.env`.
- 2. **Console Dashboard (Next.js 14 / Tailwind / Better-Auth):** The web application interface for human administrators and operators. Gated routes protect sensitive credentials. Configured via `apps/console/.env`.
- 3. **Voice Agent Worker (Python 3.11 / LiveKit Agents):** High-performance media daemon maintaining sub-200ms WebRTC pipelines, voice activity detection (VAD), speech-to-text, LLM inference, and voice streaming. Configured via `apps/ai/.env`.

1.2 Core Security Secrets (Mandatory Step 0)

Environment Variable	File Target	Security Specification	Key Purpose
SECRET_ENCRYPTION_KEY	apps/server/.env	64 Hex characters (32 bytes random)	AES-256-GCM encryption of third-party API keys and carrier SIP credentials at rest.
INTERNAL_API_KEY	apps/server/.env & apps/ai/.env	High-entropy base64 string (32+ chars)	Mutual authentication between Node API and Python AI worker; prevents unauthorized call injection.
BETTER_AUTH_SECRET	apps/server/.env	High-entropy secret string	Cryptographic signing of user authentication cookies and session bearer tokens.
LIVEKIT_API_KEY / LIVEKIT_API_SECRET	Both server & worker	LiveKit Cloud project credentials	Room creation, WebRTC token issuance, and SIP ingress/egress dispatch control.

Zero-Tolerance Security Rule: Admin Account Only

All AI provider keys (OpenAI, Anthropic, DeepSeek, Sarvam AI, Deepgram, ElevenLabs) and Telephony SIP credentials (Vobiz, Twilio, Telnyx) represent critical financial liabilities. In QuickVoice, the **Provider & API Configuration** dashboard (`/settings/providers`) is cryptographically and logically restricted to **Administrator** and **Owner** roles only. Regular organization members cannot view, alter, or extract these keys.

2. Dedicated Provider & API Management (Admin Account Only)

Administrators configure and manage all carrier and model credentials directly in the QuickVoice Console UI under **Settings → AI & Telephony APIs** (`/settings/providers`). The dashboard is segregated into four dedicated operational domains:

2.1 Tab 1: Telephony & Elastic SIP Trunks TELEPHONY & SIP

QuickVoice supports hybrid carrier integration, enabling concurrent inbound and outbound routing across Vobiz, Twilio, Telnyx, and native LiveKit SIP dispatchers.

Vobiz Cloud Telephony & Elastic SIP Trunking

Vobiz is an AI-first cloud telephony provider specifically tailored for modern real-time voice applications and India regulatory compliance (TRAI 140/92 series). Key technical parameters:

- **SIP Domain Proxy:** `sip.vobiz.ai` (Port 5060 UDP/TCP, Port 5061 TLS)
- **Authentication Mode:** SIP Digest Authentication (Username + Auth Token) or IP Access Control List (ACL) whitelisting.
- **Codec Negotiation:** G.711u (PCMU), G.711a (PCMA), and Opus (48kHz full-band for pristine AI audio).
- **Indian DID Number Allocations:**
 - **140 Series:** Dedicated transactional telemarketing numbers for India financial & enterprise outbound calling.
 - **92 Series:** Commercial business customer service and verified conversational calling.
 - **1800 Series:** National toll-free customer support lines.
 - **Standard E.164:** Full international routing (e.g., `+91xxxxxxxxxx` for India, `+1xxxxxxxxxx` for North America).

- **LiveKit Cloud SIP Inbound Dispatch Setup:**

```
# In LiveKit Cloud Console -> SIP Inbound Trunks:
Name: Vobiz-India-Inbound
SIP URI: sip:inbound.livekit.cloud
Numbers Accepted: +91140XXXXXXX, +9192XXXXXXX
Auth: IP Whitelist -> Add Vobiz Signaling IP Blocks
Dispatch Rule: Direct to Agent Room ("quickvoice-inbound-{call_id}")
```

- **LiveKit Cloud SIP Outbound Trunk Setup:**

```
# In LiveKit Cloud Console -> SIP Outbound Trunks:
Name: Vobiz-Outbound-Trunk
Address: sip.vobiz.ai:5060
Transport: TLS (Recommended) or UDP
Credentials: Vobiz Account Username & Auth Token
Caller ID: +91140XXXXXXX (Verified Vobiz DID)
```

Twilio & Telnyx SIP Credentials

Provider	Required Settings in Console	Use Case & Geography
Vobiz	API Key, Auth Token, SIP Domain (<code>sip.vobiz.ai</code>), Outbound Trunk ID	India domestic (140/92/1800), Asia-Pacific high-throughput low-latency SIP.
Twilio	Account SID, Auth Token, Elastic SIP Trunk SID (<code>TKxxxxxxxx</code>)	North America & European toll-free, global local number inventory.
Telnyx	API v2 Token, SIP Connection ID	Ultra-low per-minute carrier pricing, multi-region private IP termination.
LiveKit SIP	Inbound SIP Trunk ID, Outbound SIP Trunk ID	Native WebRTC-to-PSTN bridging with automated room dispatching.

2.2 Tab 2: Speech-to-Text (STT) Engines SPEECH-TO-TEXT

QuickVoice provisions industry-leading streaming STT providers to transcribe customer audio in under 300 milliseconds.

Sarvam AI — Vernacular Speech Recognition (Saaras v3)

- **Model Identifier:** sarvam/saaras:v3
- **Specialization:** State-of-the-art acoustic modeling for Indian languages with code-mixed vernacular understanding (Hinglish, Tanglish, etc.).
- **Supported Languages:** Hindi, Tamil, Telugu, Kannada, Bengali, Marathi, Gujarati, Malayalam, Punjabi, Odia, and Indian English (en-IN).
- **Configuration in Console:** Enter Sarvam AI API Key under Settings → AI & Telephony APIs → Speech-to-Text.

Deepgram Nova-3 & Nova-2

- **Model Identifiers:** deepgram/nova-3 (Default) and deepgram/nova-2
- **Performance:** Sub-250ms streaming transcription with smart formatting, automatic numeral conversion, profanity filtering, and keyword boosting.

2.3 Tab 3: Voice Synthesis & TTS Engines VOICE & TTS

Deliver life-like, expressive voice responses matching your brand's regional and stylistic persona.

TTS Provider	Engine Model	Key Personas & Accents	Latency Profile
Sarvam AI	sarvam/bulbul:v3	Indian Voices: <i>Shubh</i> (Professional Male), <i>Meera</i> (Warm Female), <i>Dhruv</i> (Energetic Male), <i>Ananya</i> (Conversational Female), <i>Aditya</i> (Executive Male).	~280ms streaming TTFB. Native Hindi, Tamil, Telugu, English (IN).
Deepgram	deepgram/aura-2	English & Global: <i>Asteria</i> (Energetic), <i>Apollo</i> (Confident), <i>Hera</i> (Warm), <i>Zeus</i> (Deep Baritone), <i>Luna</i> (Gentle).	~180ms instant streaming.
ElevenLabs	elevenlabs/eleven_flash_v2_5	Studio-grade emotional expressiveness, custom voice cloning, multi-dialect support.	~300ms. Premium VIP queues.
Cartesia	cartesia/sonic	High-speed conversational synthesis with realistic breathing and turn-taking.	<150ms ultra-low latency.

2.4 Tab 4: Cloud LLM Intelligence Matrix CLOUD LLMs

QuickVoice provides unified support for all premier foundation models. Administrators enter API keys once, and all models become instantly selectable across every conversational agent in the organization.

Provider Category	Models Available in QuickVoice	Best Fit / Recommended Use Case
OpenRouter (Aggregator)	openrouter/deepseek/deepseek-r1 openrouter/deepseek/deepseek-chat openrouter/meta-llama/llama-3.3-70b-instruct openrouter/qwen/qwen-2.5-72b-instruct	Cost-effective open weights, high rate limits, complex conversational reasoning with DeepSeek R1.
DeepSeek Direct	deepseek/deepseek-chat (V3) deepseek/deepseek-reasoner (R1)	Direct high-speed API access, enterprise prompt caching, ultra-low token pricing.
OpenAI	openai/gpt-4o openai/gpt-4o-mini openai/o3-mini openai/o1	High conversational fidelity, complex function calling, multi-step structured data extraction.
Anthropic Claude	anthropic/claude-3-7-sonnet anthropic/claude-3-5-sonnet anthropic/claude-3-5-haiku	Industry benchmark for conversational naturalness, zero-shot nuance, and human empathy.
Google Gemini	google/gemini-2.0-flash google/gemini-1.5-flash google/gemini-1.5-pro	Blazing fast real-time TTFT (<200ms), 1M+ token context windows, audio multimodal integration.
Amazon Bedrock	bedrock/us.anthropic.claude-3-5-sonnet bedrock/us.anthropic.claude-haiku-4-5	Enterprise AWS VPC isolation, HIPAA/SOC2 compliance, IAM role-based authentication.

3. Connecting LiveKit Cloud & Media Infrastructure

3.1 LiveKit Cloud Signaling & Room Orchestration

LiveKit Cloud powers the real-time WebRTC media mesh connecting callers, carriers, and the QuickVoice AI worker.

1. Navigate to <https://cloud.livekit.io> and create a production project.
2. Under **Project Settings** → **Keys**, generate an API Key and Secret.
3. Insert the credentials into `apps/server/.env` and `apps/ai/.env`:

```
LIVEKIT_URL=wss://your-project.livekit.cloud
LIVEKIT_API_KEY=APIxxxxxxxxxxxxxx
LIVEKIT_API_SECRET=secretxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

4. Deploy the QuickVoice AI worker: `python -m apps.ai.main start` (or via Docker Compose: `docker-compose up -d ai-worker`). The worker immediately registers itself as an active participant dispatcher on LiveKit Cloud.

3.2 LiveKit SIP Dispatch Rule Configuration

When an inbound PSTN call arrives from Vobiz, Twilio, or Telnyx, LiveKit dispatches it into an isolated WebRTC audio room. Configure the SIP dispatch rule via LiveKit CLI or Console:

```
# livekit-cli sip dispatch-rule create rule.json
{
  "name": "QuickVoice-Auto-Agent-Dispatch",
  "rule": {
    "dispatchRuleIndividual": {
      "roomPrefix": "call-"
    }
  },
  "metadata": "{\"source\": \"vobiz_twilio_gateway\"}"
}
```

4. Inbound & Outbound Webhook Architecture

4.1 Webhook Lifecycle Events

QuickVoice dispatches cryptographic webhooks to third-party CRMs (HubSpot, Salesforce, Zoho, Zapier) at key lifecycle transitions:

Event Name	Trigger Moment	Payload Key Attributes
<code>call.started</code>	SIP INVITE answered; WebRTC room joined.	<code>callId</code> , <code>callerNumber</code> , <code>agentId</code> , <code>direction</code> , <code>timestamp</code>
<code>call.transcript.chunk</code>	Real-time streaming speech chunk committed.	<code>callId</code> , <code>speaker</code> (<code>agent/user</code>), <code>text</code> , <code>confidence</code>
<code>tool.invoked</code>	LLM triggers external tool (e.g. Cal.com booking).	<code>toolName</code> , <code>inputArguments</code> , <code>executionStatus</code>
<code>call.ended</code>	SIP BYE received; room closed.	<code>durationSeconds</code> , <code>disposition</code> , <code>recordingUrl</code>
<code>call.analyzed</code>	Post-call AI summary & sentiment completed.	<code>summary</code> , <code>sentiment</code> , <code>extractedData</code> , <code>costUsd</code>

4.2 HMAC-SHA256 Signature Verification

Every webhook request includes an `X-QuickVoice-Signature` header. Verify signatures in Node.js/Python to prevent forgery:

```
// Example Node.js Webhook Receiver
const crypto = require("crypto");

function verifyQuickVoiceWebhook(req, secret) {
  const signature = req.headers["x-quickvoice-signature"];
  const hmac = crypto.createHmac("sha256", secret);
  const digest = hmac.update(JSON.stringify(req.body)).digest("hex");
  return crypto.timingSafeEqual(Buffer.from(signature), Buffer.from(digest));
}
```

4.3 Custom CRM API Tools: Customer Ticket Generation & Live Querying

QuickVoice voice agents can dynamically collect caller information, invoke external REST APIs mid-conversation, push data to your CRM server, and receive real-time answers to read back to the customer.

How the CRM Ticket Lifecycle Operates in Real-Time Voice:

1. **Conversation & Extraction:** When configured with parameters marked `valueType: "LLM Prompt"` (e.g. `customer_name`, `email`, `issue_description`), the voice agent naturally asks the caller for these details during the phone call.
2. **HTTP Execution:** Once the necessary fields are collected, the AI worker triggers an internal HTTP dispatch to your CRM endpoint (e.g. `POST https://api.mycrm.com/v1/tickets`) with headers and authorization tokens securely injected from organization secrets.
3. **CRM Response Parsing:** Your CRM generates the ticket and responds with a JSON payload (e.g. `{ "ticket_id": "TICK-9428", "status": "ASSIGNED", "sla_hours": 4 }`).
4. **Spoken Confirmation:** The voice agent immediately consumes the returned JSON and confirms to the caller: *"Thank you, Sarah! I have created ticket number TICK-9428 for your order issue. Our tier-1 support will follow up with you within 4 hours."*
5. **Bidirectional Querying (GET):** If a customer calls asking *"What is the status of ticket 10482?"*, an attached lookup tool (`GET /v1/tickets/{ticket_id}`) queries the CRM and the agent reads the current status and latest agent notes directly to the caller.

4.4 Uploading & Importing APIs in JSON Format

Administrators do not need to manually configure parameters one-by-one. QuickVoice provides a dedicated **"Import JSON API"** modal in the Console UI (under **Tools** → **Import JSON API** or inside the **New Tool** sheet):

Supported JSON Specification Formats:

- **QuickVoice Native JSON Specification:**

```
{
  "name": "CRM Generate Ticket",
  "description": "Creates customer support ticket in company CRM",
  "api_url": "https://api.mycrm.com/v1/tickets",
  "api_method": "POST",
  "api_headers": [
    { "key": "Authorization", "value": "Bearer YOUR_CRM_SECRET", "type": "Secret" }
  ],
  "api_body": [
    { "name": "customer_name", "type": "String", "valueType": "LLM Prompt", "required": true },
    { "name": "customer_email", "type": "String", "valueType": "LLM Prompt", "required": true },
    { "name": "issue_description", "type": "String", "valueType": "LLM Prompt", "required": true },
    { "name": "priority", "type": "String", "valueType": "Static Value", "value": "HIGH" }
  ]
}
```

- **Postman / REST Spec:** Accepts standard requests with `url`, `method`, `headers`, and sample `body` payloads. Parameters are automatically converted into LLM function-calling schemas.
- **1-Click Built-In CRM Templates:** One-click load buttons for *"CRM Generate Ticket (POST)"* and *"CRM Ticket Status Lookup (GET)"*.

4.5 Interactive Tool Testing & Live Response Inspector

Before attaching a tool to live agents, administrators can verify their CRM endpoint directly inside the QuickVoice Console:

1. Click the **"Test"** button on any tool card or inside the tool editor.
2. The **Tool Test Modal** automatically generates input fields for each caller parameter.
3. Click **"Sample Data"** to auto-fill realistic customer ticket values (name, email, phone, issue).
4. Click **"Send Test Request"**. QuickVoice executes the request from the server, enforces URL safety, and returns:
 - **HTTP Status Badge:** Confirms `200 OK` or `201 Created`.
 - **Latency Monitor:** Displays exact response round-trip time in milliseconds (e.g. ⚡ `142ms`).
 - **JSON Response Inspector:** Formatted syntax-highlighted JSON viewer with a 1-click Copy button.
 - **Voice Agent Simulation:** Previews how the LLM will parse the returned attributes and speak the confirmation to the caller.

5. Step-by-Step Agent Creation & Model Selection via Console UI

All agent creation, prompt engineering, speech engine configuration, and carrier number binding can be performed entirely within the QuickVoice Console UI by administrators.

1 Log in with Administrator Privileges

Navigate to `http://localhost:3005/login` and sign in using your Admin account (e.g., `admin@quickvoice.ai`). Non-admin accounts will not see provider API tabs or permission matrices.

2 Update Provider API Keys (Settings → AI & Telephony APIs)

Before launching agents, verify your provider credentials under **Settings → AI & Telephony APIs**:

- **Telephony:** Enter your Vobiz API Key, Auth Token, and SIP Domain (`sip.vobiz.ai`).
- **STT:** Enter Sarvam AI API Key (for Indian regional languages) and Deepgram API Key.
- **TTS:** Select Sarvam Bulbul voice or Deepgram Aura.
- **LLM:** Enter keys for OpenRouter (DeepSeek R1/V3), OpenAI, Anthropic, or Google Gemini.
- Click **Save All Provider APIs**. Settings are securely encrypted with AES-256 and persisted in organization metadata.

3 Create a New Agent Entity

In the main navigation, click **Agents** and press the **+ New Agent** button. Enter the agent's display name (e.g., *"Support Concierge - India"*) and choose a starter template or blank canvas.

4 Configure Agent Behavior & LLM (Behavior Tab)

In the Agent Detail view, open the **Behavior** tab:

- **Greeting / First Message:** Set the initial spoken greeting (e.g., *"Namaste! Thank you for calling QuickVoice support. How may I assist you today?"*).
- **System Prompt:** Define instructions, rules, boundary conditions, and conversational persona.
- **LLM Model Selection:** Choose from the newly expanded catalog: `openrouter/deepseek/deepseek-r1` , `anthropic/claude-3-7-sonnet` , `openai/gpt-4o` , OR `google/gemini-2.0-flash` .
- **Temperature:** Adjust between 0.1 (strict procedural adherence) and 0.7 (creative, conversational).

5 Configure Speech & Regional Persona (Voice Tab)

Open the **Voice** tab to select STT and TTS models:

- **Speech-to-Text:** Choose `sarvam/saaras:v3` for Hindi/Indian vernacular or `deepgram/nova-3` for global English/multilingual.
- **Voice (TTS) Selection:**
 - For Indian regional accents: Select **Sarvam Shubh** (Male) or **Sarvam Meera** (Female).
 - For global personas: Select **Deepgram Asteria**, **Hera**, or **ElevenLabs Flash**.
- **Audition Audio:** Click **Play Sample Audio** in the Voice Profile Panel to test cadence, pitch, and inflection.

6 Assign Telephony Number (Numbers Tab)

Navigate to **Phone Numbers** in the console. Select a purchased or imported number (e.g. Vobiz 140/92 DID) and click **Assign Agent**. Inbound calls to this number will immediately spin up the configured agent voice pipeline.

6. Verification, System Health & Diagnostics

6.1 Automated Health Check Suite

Execute the following diagnostic commands from any authorized administrative workstation:

Service Component	Diagnostic Command	Expected Response
API Server Health	<code>curl http://localhost:5000/api/v1/system/readiness</code>	<code>{"status":"ready","database":true,"redis":true}</code>
Console Web Interface	<code>curl -I http://localhost:3005/settings/providers</code>	HTTP/1.1 200 OK (or 307 redirect if unauthenticated)
Swagger API Documentation	<code>curl -I http://localhost:5000/api/v1/docs</code>	HTTP/1.1 200 OK
Python Voice Worker	<code>curl http://localhost:5555/health</code>	<code>{"status":"healthy","livekit":true,"stt":"ready","tts":"ready"}</code>

6.2 Administrator Troubleshooting Matrix

Symptom / Error Code	Underlying Root Cause	Mandatory Administrative Remediation
403 Forbidden on /settings/providers	User account possesses standard member role instead of admin or owner .	Promote user role via Settings → Roles or update membership directly in PostgreSQL.
Vobiz SIP 401 Unauthorized	Mismatched SIP Auth Token or IP address not whitelisted in Vobiz Portal.	Verify credentials in /settings/providers ; add LiveKit egress IP to Vobiz ACL.
Sarvam AI 400 Bad Request	Unsupported language code supplied to bulbul:v3 .	Ensure language is set to supported Indian ISO code (hi , ta , te , kn , bn , en-IN).
OpenRouter 402 Payment Required	Depleted credits on OpenRouter account for DeepSeek R1 / V3.	Top up OpenRouter balance at openrouter.ai/credits .
STT Latency Spike (>1000ms)	AI worker container CPU throttling or network jitter to STT endpoints.	Allocate minimum 2 vCPU per 10 concurrent streams; verify low-latency DNS resolution.
listen EADDRINUSE :3000	Port 3000 occupied by existing container or service.	QuickVoice Console is pre-configured on Port 3005 (next dev -p 3005).

Enterprise Support & Production Repository

Source code, automated migration scripts, and architecture blueprints are maintained at:

<https://github.com/mithunccdev/Advanced-quick-voice.git>